

Supplementary Information: First-order conflation and recurrent fragmentation generate a cognitive near-singularity

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Contents and status

This Supplement is the standalone formal, computational and status record for Article v15. It restores every load-bearing definition and premise used by the Article, separates semantic registration from proof checking, retains the executed finite realization from the earlier computational package and records all unexecuted gates explicitly. No unpublished artifact is required to recover the formal content.

Record fields: Package · Content · v15 status

A — Content: Core FMI-CNS mathematics; v15 status: Prose proofs and countermodels supplied; kernel formalization pending
B — Content: Endogenous semantic geometry; v15 status: Definitions and theorem targets supplied; physical mapping remains a hypothesis
C — Content: Lean verification artifact; v15 status: Module and declaration manifest supplied; pinned build and axiom audit pending
D — Content: String-specific extension; v15 status: Candidate geometric representation; comparative utility experiment pending
E — Content: External and propagation evidence; v15 status: Retained synthetic witness; receiver, external trace, branch and hazard studies open

The current status tuple is therefore

$$\text{Status}(v15) = (\text{conditionally closed}, \text{conditionally closed}, \text{open}, \text{open}, \text{open}).$$

Formal coherence within stated premises and submission-carrier readiness are separate verdicts.

Supplementary Note 1. Typed metric functional state spaces and endogenous semantics

Local map. This Note defines the mathematical object on which every later theorem operates. Its payoff is a representation of semantic identity that is constituted by metric position, typed interactions and admissible relational configuration rather than by a prose label.

Definition S1: typed metric FSS

A typed metric functional state space is

$$X = (V, E, \tau_V, \tau_E, g, \pi, \beta, A),$$

where V is the set of represented functional states, E the set of typed interaction or transition instances, τ_V and τ_E the node and interaction typing maps, g a metric or metric family, π an optional coordinate or embedding representation, β the boundary record and A the family of admissible transformations and trajectories.

A boundary record may contain domain and codomain restrictions, certificates, provenance, residuals, successor obligations, fitness relations, continuation conditions, version identities and target scope. The metric may be primitive, interaction-induced or jointly defined. Every theorem must state which case it uses.

Definition S2: nodes, processes and functions

A represented concept or functional state is a node $v \in V$. A reasoning or physical interaction is a typed edge, hyperedge, path or process subgraph. Definitions, claims, functions, proofs, experiments, datasets and manuscript passages are semantic roles borne by nodes and processes rather than additional primitive represented-object kinds.

A represented function has an intensional node carrying identity, domain, codomain and contract, and an extensional family of process instances carrying executions. An instantiates relation connects the contract node to those instances.

Definition S3: semantic profile and identity

For $v \in V$,

$$Prof_X(v) = (X, v), Sem_X(v) = [(X, v)]_{\cong_A}.$$

An admissible pointed isomorphism $F: (X, v) \rightarrow (Y, w)$ must satisfy:

$$F(v) = w,$$

preserve node and interaction types, incidence, orientation and the declared metric relations, preserve boundary, certificate, warrant and continuation records, and intertwine the relevant transition and participant-fitness relations. Approximate isomorphisms must expose the tolerance and residual rather than silently replacing equality.

Theorem S1: semantic equivalence

If $F: (X, v) \rightarrow (Y, w)$ is an admissible target-preserving pointed isomorphism, then every invariant target proposition expressible in the registered language has the same truth value at v and w .

Proof. Let the registered language be generated from atomic predicates on node type, edge type, metric relation, incidence, boundary, warrant, transition, fitness and continuation structure by Boolean connectives and bounded quantification over the represented FSS. Admissibility preserves every atomic predicate. The result follows by induction on formula construction. For a quantified formula, the bijection induced by F transports witnesses and counterexamples. Therefore truth is invariant. QED.

Definition S4: finite target-complete semantic kernel

For target family T and resource budget B , define

$$K_T(v; X, B) = \min \{ K \subseteq X : T \text{ is decidable for } v \text{ from } K \text{ within } B \}.$$

If no unique minimum exists, retain the family of minimal kernels. The finite semantic signature is

$$Sem_{T,B}(v) = [K_T(v; X, B), v]_{\cong_{T,A}}.$$

Failure to certify a target-complete finite kernel yields undetermined, not a fabricated complete signature.

Cross-FSS structural principle

Conceptual space and physical P-space instantiate the same invariant-level principle: functional identity is constituted by metric position, typed interactions and admissible relational configuration. The operands and substrates remain distinct. This is a structural identity claim, not object identity.

Interactions induce topology

For each operation o , define an admissible-transition relation $R_o \subseteq V \times V$. The family $\{R_o\}$ induces or constrains reachability, connectedness, support overlap, commutation classes, boundary crossings and path-equivalence structure. Interactions therefore generate topological implications; topology is not merely a label applied after the dynamics are fixed.

Supplementary Note 2. Reflective FMI subspace and signal boundaries

Local map. This Note identifies the reflective object needed to register the complete argument and separates signal carriers from internal conceptual states. It supplies the transmission residuals used by governance and propagation theorems.

Theorem S2: reflective-subspace representation

Let C be a conceptual FSS. The reflective FMI subspace $R_{FMI} \subseteq C$ contains represented concepts, process identities, function contracts, compositions, targets, proofs, experiments, warrants, governance rules, signal residuals and unresolved external premises.

Every registered object required to state, prove, test, amend or transmit the FMI-CNS-Great Filter relationship can be represented as a concept node, a process edge or subgraph, or both. No additional primitive represented-object class is required.

Proof. Map every noun-like identity or contract to a concept node. Map every transformation, inference, execution, comparison, registration, transmission or amendment to a typed process. Represent multi-input operations as hyperedges or contract-preserving subgraphs. Represent a proof as a premise-to-conclusion subgraph with certificate; represent an experiment as protocol, execution, measurement and interpretation subgraphs. This construction covers every registered role. A counterexample must exhibit an indispensable represented object that cannot be encoded in these forms without changing its contract. QED.

Reflective strata are typing devices. Reifying a process as a concept improves observability but does not increase cognitive order unless it enables another non-equivalent space-spanning composition.

Signal boundary

For participant i ,

$$Register_i: \Sigma_i \rightarrow X_i, Express_i: X_i \rightarrow \Sigma_i.$$

A signal is not an internal concept until it is registered. An internal concept is not externally available until it is expressed.

Four residuals

For a composition executed by i and reconstructed by j ,

$$R_{i \rightarrow j} = R_{reg,i} \oplus R_{expr,i} \oplus R_{recv,j} \oplus R_{recon,j}.$$

The direct sum preserves failure source and type. $R_{reg,i}$ is source structure unavailable after initial registration; $R_{expr,i}$ is structure omitted or altered by expression; $R_{recv,j}$ is expressed structure not registered by the receiver; and $R_{recon,j}$ is registered structure not reconstructed into an executable contract.

Registration closure

A formalization is registration-closed for target family T when every target-relevant semantic invariant in the manuscript has a canonical mathematical representation and every manuscript use maps to that representation without unresolved ambiguity. Registration closure is distinct from proof closure: a proof assistant can validate a proof about the wrong registered object.

Supplementary Note 3. FMI function contracts and four-operation topology

Local map. This Note states the canonical FMI function registry and replaces the undefined local/global terminology of v14 with target-relevant factorization definitions. It also supplies the type obstruction needed for Register/Recall nonreduction.

FMI registry

Record fields: ID · Function · Layer · Contract

B01 — Function: Register; Layer: boundary; Contract: Admit typed external signal structure without minting unavailable distinctions
B02 — Function: Express; Layer: boundary; Contract: Encode selected FSS structure onto a carrier while recording loss
C01 — Function: Store; Layer: conceptual; Contract: Preserve recoverable identity, relation, verdict, obligations and provenance
C02 — Function: Recall; Layer: conceptual; Contract: Reactivate or reconstruct stored structure under target and budget
C03 — Function: L; Layer: conceptual; Contract: Evaluate or transform within active local support
C04 — Function: G; Layer: conceptual; Contract: Evaluate or transform across globally relevant support while preserving obligations
C05 — Function: Promotion/Create; Layer: conceptual; Contract: Preserve an admissible composite as a reusable higher-order operand
E01 — Function: Choice; Layer: external control; Contract: Select the next external-function invocation through the fitness-control basis
F01-F06 — Function: Model, Evaluate, Stability, Adaptation, Decomposition, Bridging; Layer: fitness; Contract: Represent, compare, preserve, correct, factor and bridge participant consequences
X01 — Function: Certificate; Layer: coupling; Contract: Validate target-relevant paths and boundaries
X02 — Function: Governance; Layer: coupling; Contract: Authorize participant-complete selection and target revision

For current, target and projected fitness,

$$\chi(a) = (f_{tar} - f_{cur}, f_{proj}(a) - f_{cur}, f_{tar} - f_{proj}(a)).$$

The current-projected relation grounds Model/Evaluate, current-target grounds Stability/Adaptation, and target-projected grounds Decomposition/Bridging. A non-embeddable partial order remains partial and yields undetermined.

Support definition

For operation instance o and target Φ ,

$$Supp_{\Phi}(o) = \cap \{ S \subseteq V : \text{the output and retained invariants of } o \text{ are determined from } S \}.$$

When the intersection is not unique, retain the family of minimal supports.

An operation has **local support** when every minimal target support lies within one declared interaction neighbourhood and factorizes over target-independent remote contexts. It has **global support** when some target-relevant output or invariant depends nonfactorizably on extended support spanning multiple neighbourhoods or components. Global does not mean acausal or spatially infinite; local does not mean short-range. The distinction is target-relevant factorization.

Schedule definition

For admissible local realizations r_1, r_2 , the schedule is **parallel** when

$$r_1 \circ r_2 \cong_T r_2 \circ r_1,$$

and **sequential** when a typed precedence relation is load-bearing and reversal changes the target-relevant result or makes execution inadmissible. Parallel means observational commutation, not simultaneous wall-clock execution.

Candidate four-operation inventory

Record fields: Operation · Support · Schedule · Cognitive role · Candidate physical realization

Register — Support: local; Schedule: parallel; Cognitive role: constrained placement or binding; Candidate physical realization: strong interaction

Recall — Support: global; Schedule: sequential; Cognitive role: globally constrained typed reconfiguration; Candidate physical realization: weak interaction

L — Support: local; Schedule: sequential; Cognitive role: ordered local propagation or calibration; Candidate physical realization: electromagnetic interaction

G — Support: global; Schedule: parallel; Cognitive role: extended-support geometry-sensitive coherence; Candidate physical realization: gravity

The operation classes are defined before the physical assignments. Each physical assignment remains a separately testable mapping hypothesis.

Theorem S3: Register/Recall nonreduction

Let $\langle L, G \rangle$ be the typed algebra generated by compositions of L and G acting as endomorphisms on active conceptual state X . Register has type $\Sigma \rightarrow X$ and Recall has type $M \times T \times B \rightarrow X$.

No well-typed term generated solely by L and G has the boundary type of Register or the memory-access type of Recall. A reduction requires an additional bridge implementing the missing boundary or memory contract and therefore does not establish generation by L and G alone.

Proof. Every generator has domain and codomain X under the base architecture; composition preserves this endomorphism type. Register requires a source in Σ , while Recall requires access to $M \times T \times B$. Neither source is constructible by endomorphism composition without a typed bridge. Adding such a bridge supplies the missing primitive contract. QED.

Supplementary Note 4. Projection, conflation and reflective necessity

Local map. This Note proves the information-theoretic obstruction that begins the formal chain and records countermodels that delimit it.

Let $q_{FO} = h \circ q_R$. Every target factoring through q_{FO} also factors through q_R . Some targets factor through q_R but not q_{FO} when h removes target-relevant distinctions.

Theorem S4: first-order conflation

If $q_{FO}(x) = q_{FO}(y)$ but $\Phi(x) \neq \Phi(y)$, no evaluator restricted to the projected state is correct on both. The proof is the contradiction argument in the Article. For stochastic evaluators with equal registered states and kernels, the output laws coincide.

Corollary S4.1: local stability

For deterministic first-order-local L and equal inputs,

$$L^k(q_{FO}(x)) = L^k(q_{FO}(y))$$

for every finite k . Induction on k proves the result.

Theorem S5: reflective necessity

When a target depends on function identity, contract, composition, provenance, warrant, participant consequences or successor obligations, a representation containing only domain concepts and conclusions is inadequate whenever it collapses two process structures with different target verdicts. This is T-CONFLATE applied to a target whose separating witness belongs to the reflective subspace.

Countermodels

Record fields: Countermodel · Effect

Target fully determined by first-order projection — Effect: Reflection unnecessary for that target

Reflective registry without new composition — Effect: Observability improves; order does not increase

Equivalent renaming or call expansion — Effect: No new composition class

Projection retains every target-relevant distinction — Effect: First-order assessment can be exact

Target-aware abstention — Effect: False confidence falls without increasing span

Supplementary Note 5. Composition equivalence, span, order and invariance

Local map. This Note distinguishes intrinsic cognitive order from finite evidence about order. It carries the representation-invariance proof and demotes the cardinality increment to a lemma.

Definitions

A registered composition $\Gamma = f_m \circ \dots \circ f_1$ has identity, signature, domain, codomain, boundary conditions, execution semantics and preserved invariants.

For admissible domain D and target language L ,

$$\Gamma_1 \simeq_{D,L} \Gamma_2 \Leftrightarrow \forall x \in D, \text{Obs}_L(\Gamma_1 x) = \text{Obs}_L(\Gamma_2 x).$$

A composition spans the declared universal target family $T^{\hat{c}}$ when it connects every target-required region by certified transformations, preserves semantic identities and obligations, and returns correct or calibrated-undetermined verdicts within the architecture's contract.

$$S(C) = \{[\Gamma]_{\simeq} : \Gamma \text{ registered, reusable, nonredundant and spanning}\}, \text{Ord}(C) = |S(C)|.$$

A finite certificate is

$$\widehat{\text{Ord}}_{T,\epsilon,B}(C) = |\{[\Gamma]_{\simeq} : \text{SpanCert}_{T,\epsilon,B}(\Gamma, C)\}|.$$

It is a lower bound unless target and certificate completeness are proved.

Theorem S6: representation invariance

If $F : C \rightarrow C'$ is an admissible FSS isomorphism preserving external-function identity, composition, span, reuse and functional equivalence, then $\text{Ord}(C) = \text{Ord}(C')$.

Proof. Map each class $[\Gamma] \in S(C)$ to $[F \Gamma F^{-1}] \in S(C')$. Preservation assumptions make this map well defined. The inverse is induced by F^{-1} , so the map is bijective. Cardinalities are equal. QED.

Lemma S6.1: order increment

If a new qualifying class $[\Gamma_{N+1}]$ is not already in $S(C_N)$, then the union with that singleton increases cardinality by one. This is definitional and does not independently establish reversal or operand expansion.

Finite decidability target

For finite FSSs, finite operation libraries, decidable target predicates and bounded resources, span certification and functional equivalence are decidable by exhaustive enumeration. The executable Lean decision theorem remains an implementation target.

Supplementary Note 6. Arrival-service fragmentation, reversal and recurrence

Local map. This Note replaces the v14 directional fragmentation tendency with two explicit theorem levels: backlog divergence and regional fragmentation. It then proves reversal and recurrence under stated service conditions.

Let $D_N(t)$ be cumulative target-weighted arrivals, $B_N(t)$ cumulative certified service and

$$U_N(t) = \max\{0, D_N(t) - B_N(t)\}.$$

Theorem S7: backlog divergence

If $D_N(t) - B_N(t)$ is unbounded above, then $U_N(t)$ is unbounded along a subsequence. If the limsup linear drift is positive, the difference is unbounded. The proof follows directly from the maximum definition.

Theorem S8: regional fragmentation

Construct an obligation graph H_t . Assume local integration keeps each region below ϵ_{local} , cross-region arrivals exceed certified cross-region service by an unbounded amount, unresolved edges are not replaced

by false identity certificates and the target requires at least two regions. Then an unbounded subsequence contains an unresolved target-required cross-region edge while each incident region remains internally certified. QED.

Theorem S9: reflective-loss burden

Let C_t be target-different pairs collapsed by the first-order projection and let

$$Q_t = \sum_{(x,y) \in C_t} w_t(x,y).$$

If Q_t is unbounded and no additional separating distinctions are registered, the minimum first-order decision loss or calibrated-undetermined mass grows according to the declared loss. The exact lower bound depends on the prior and loss function and must be reported with them.

Theorem S10: reversal

If an added composition changes service so that $D_N(t) - B_{N+1}(t) \leq \epsilon$ over a declared interval, then $U_{N+1}(t) \leq \max(0, \epsilon)$ over that interval. The current increasing-backlog regime is reversed to bounded residual.

Theorem S11: recurrence

If the lift creates a new arrival process D_{N+1} and $D_{N+1}(t) - B_{N+1}(t)$ becomes unbounded above without another spanning composition or scaling change, then $U_{N+1}(t)$ is unbounded along a subsequence by Theorem S7.

Required countermodels

Bounded domains, matching same-order service, target families ignoring the residual, distributed service that grows faster than arrivals, temporary backlog decreases, reflection-only improvements and false nominal bridges are all retained as explicit theorem boundaries.

Supplementary Note 7. CNS lift, near-singular content and dynamical bridge

Local map. This Note places the scientific burden beyond the definitional count change. It connects the order-lift transition to operand expansion, reversal and the separately published dynamical CNS.

Theorem S12: order-lift CNS

If an order- N architecture gains a registered, reusable and non-equivalent spanning composition, expands the admissible operand domain and closes the current frontier within its residual contract, then order becomes $N+1$, the operand domain expands, the current backlog is bounded over the declared interval and complete consequences need not factor through the order- N projection.

Proof. Lemma S6.1 gives the order increment. Operand expansion is a premise witnessed by the new admissible substructures. Theorem S10 gives reversal. If complete consequences factored through the old projection, no new target-relevant distinction would be required; a paired target witness supplies the nonfactorization. QED.

Near-singular content

Let $AdmSub(X_N)$ be the structures that become admissible operands under the new composition. Define structural displacement

$$S_N(a) = \mu(R_N^a \{ R \dot{\downarrow} N \})$$

and participant-target consequence field $\Delta F(a)$. A world comparator $B_N(X_N, W)$ bounds non-lift displacement and a lift lower bound $L_N(X_N)$ satisfies $L_N > B_N$ above the declared threshold.

Under admissible-substructure growth, the world comparator, monotone coupling, adoptability and no-order-skipping premises, a minimal lift exceeds every non-lift order- N intervention in magnitude above the threshold, while complete content need not factor through the order- N projection.

The result is conditional on the comparator and coupling; no infinite physical quantity is asserted.

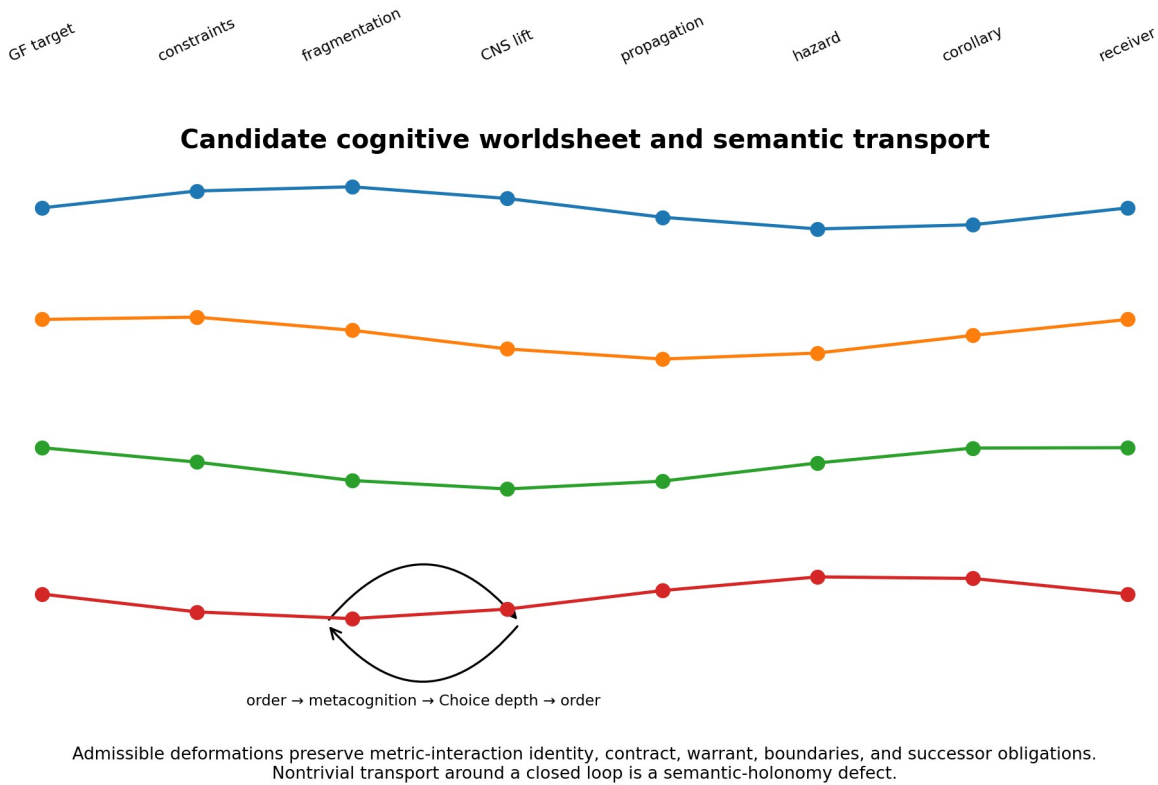
Dynamical CNS bridge

The published object is $(X, d, \otimes, G, N)$ with a globally contracting, compositionally closed and benign-reparameterization-invariant update operator converging to a unique fixed point or normal form [15].

Bridge A. If the new composition enables such an operator on the expanded domain, the order lift enters the dynamical CNS regime there.

Bridge B. If realization of a dynamical CNS operator on a previously untraversable domain requires a non-equivalent spanning composition absent at order N , it witnesses an order lift.

Countermodels include a noncontracting order lift, a contracting local operator with no new composition, a bounded-subspace fixed point, multiple compositions converging to one fixed point and a mere call expansion.



Supplementary Figure 1 | Candidate cognitive worldsheet and semantic transport.

Supplementary Note 8. Importance, governance and regenerative propagation

Local map. This Note separates magnitude from valence and defines the propagation criterion needed for collective order.

Participantwise option monotonicity

Let $O_b \subseteq O_a$. Suppose represented and realized values differ by at most ϵ_i , selection error is at most δ_i , each baseline option remains admissible and governance selects a new option only if its represented lower bound is no worse than the best baseline. Then

$$u_i(a^i) \geq \max_{b \in O_b} u_i(b) - (2\epsilon_i + \delta_i).$$

Proof. Selection gives $\hat{u}_i(a^i) \geq \max_b \hat{u}_i(b) - \delta_i$. Representation error gives $u_i(a^i) \geq \hat{u}_i(a^i) - \epsilon_i$ and $\hat{u}_i(b) \geq u_i(b) - \epsilon_i$. Combining proves the bound. QED.

If participant completeness, target stability, projection adequacy or realization fails, valence inversion is possible.

Centralized uncheckability

A projection $q: X \rightarrow Z$ is adequate for target Φ only if $q(x) = q(y)$ implies $\Phi(x) = \Phi(y)$. A centralized realization is collectively uncheckable whenever the center uses a target-relevant distinction that does not factor through the signal available to participants.

Regenerative propagation

$$\text{Regenerate}(i, j; \Gamma, T) = 1$$

only when receiver j reconstructs a contract-preserving realization, executes it on hidden targets and can retransmit it under the same criterion. $R_r(t)$ is the expected number of new regenerative executors per executor. Exposure and agreement are necessary at most; they are not regeneration.

Supplementary Note 9. Great Filter lemma family and conditional corollary

Local map. This Note derives each necessary property separately before applying the constraint set to the recurrent FMI-CNS mechanism. It prevents a list of intuitions from being presented as one theorem.

Target classes and assumptions

A terminal filter suppresses persistence of almost all technologically consequential lineages after consequential environmental transformation becomes possible. An observability filter suppresses persistent astronomically detectable manifestations. The assumptions are a many-opportunity regime, ordinary causal locality, potential branching, diverse realization and a declared residue expectation.

Lemma S13: upper-tail suppression

For independent opportunities with $p_i \geq p > 0$,

$$P(\text{no escape among } M) = \prod_i (1 - p_i) \leq (1 - p)^M \rightarrow 0.$$

A sufficient many-opportunity filter must suppress exceptional branches, correlate outcomes, reduce effective opportunities or suppress observables after escape.

Lemma S14: branch closure

For independent branches with survival probabilities $s_j > 0$,

$$P(\text{at least one survives}) = 1 - \prod_j (1 - s_j).$$

A sufficient filter acts before branching, recurs in branches, preserves a shared vulnerability, correlates outcomes or remains effective after dispersal.

Lemmas S15-S19

Causal self-instantiation: a nondistributed filter must be generated or propagated to each causally separated site before escape.

First-mover consistency: the earliest consequential lineage must be affected under the same law or a regress-closed prior condition.

Functional substrate invariance: the defining vulnerability must extend across the declared realization class rather than one contingent biology.

Adaptive non-escapability: a stable positive adaptive escape probability defeats upper-tail suppression unless adaptation is correlated or bounded.

Outcome, residue and timescale sufficiency: the target outcome must occur before durable escape or correction, or remain effective afterward; a target-specific sufficient condition can require $\tau_F < \min(\tau_D, \tau_C)$.

Theorem S20: Great Filter constraint

A sufficient member of the declared filter class must satisfy the conjunction of Lemmas S13-S19.

Theorem S21: structural admissibility

Recurrent fixed-order fragmentation and CNS reversal belong to the structurally admissible class when burden is generated locally by growth, relations are functionally substrate-general, lifts reverse current burden but recurrence recreates it, successor branches inherit or recreate the limitation and outcome/timescale premises are supplied separately.

Corollary S21.1

Let $h_t = H(U_N(t), C(t), R(t), P_N(t), X(t))$, $0 \leq h_t < 1$. If $\sum_t h_t = \infty$, then $\prod_t (1 - h_t) = 0$. If the external FSS mapping, recurrent burden, failed lift/propagation, branch correlation, non-summable hazard and target outcome premises hold, recurrent failed CNS realization or propagation is a post-intelligence Great Filter within the model.

Claim ladder

Record fields: Level · Permitted statement · Required warrant

GF-0 — Permitted statement: Some mechanisms are physically possible; Required warrant: physical consistency or example

GF-1 — Permitted statement: The declared filter class has the listed necessary properties; Required warrant: Theorem S20

GF-2 — Permitted statement: Recurrent FMI-CNS failure is structurally admissible; Required warrant: Theorem S21

GF-3 — Permitted statement: It would be a filter if mapping and hazard premises hold; Required warrant: Corollary S21.1

GF-4 — Permitted statement: Current systems exhibit measured components; Required warrant: external traces / receiver studies

GF-5 — Permitted statement: Real civilizations instantiate the full mechanism; Required warrant: physical, branch, timescale and hazard evidence

GF-6 — Permitted statement: The mechanism explains the observed Great Filter; Required warrant: comparative inference and observability assumptions

Supplementary Note 10. Cognitive string theory as semantic mathematical representation

Local map. This Note gives CST its narrowed role: it represents the semantic object that Lean can reason about. It separates the constitutive metric and hard admissibility from any optional scalar action.

Worksheet object

Let

$$X^A: \Sigma_{ws} \rightarrow C \times F$$

embed a reasoning composition and its revision family. Coordinates (σ, τ) index position along the composition and revision, alternative derivation or receiver reconstruction. Each point or cell carries a semantic state, an FMI operation label, interaction type, boundary record and warrant/certificate state.

Induced metric

The worksheet metric is induced from the ambient semantic metric:

$$h_{ab} = \partial_a X^A \partial_b X^B g_{AB},$$

or by a discrete pullback analogue for graph-based FSSs. The construction must state whether g is primitive, interaction-induced or hybrid and which transformations preserve it.

Connection and transport

A connection A_C transports concept identity, function contract, target, warrant, boundary data, successor obligations and participant roles. For path γ , transport is $PT_{A_C}(\gamma)$; for a closed loop ℓ , holonomy is $Hol_{A_C}(\ell)$.

Holonomy-invariance target. Admissibly homotopic derivations with preserved curvature and boundaries transport equivalent semantic objects.

Holonomy-drift target. If a closed derivation loop returns a target-relevant identity, contract or warrant to a non-equivalent object, it certifies semantic drift.

The discrete Lean version can use path composition and transport equality without differential geometry.

Consequence closure and hard admissibility

Let $\mathcal{E}$ be a complete lattice of candidate operand structures and $\Phi_w: \mathcal{E} \rightarrow \mathcal{E}$ a monotone consequence operator adding consequences, boundaries and successor obligations. Define

$$\mathcal{E}_w^i = \text{lfp}(\Phi_w).$$

Existence follows from the fixed-point premise. Hard admissibility is a proposition $\text{Adm}_w(\mathcal{E}_w^i)$. Missing identity, type, contract, warrant, boundary or continuation obligations are ill typed or unproved; they cannot be compensated by a lower action value.

Constitutive metric and optional action

The constitutive metric defines relative position, neighbourhood, admissible isometry, deformation size and any required contraction. An optional action can rank or search admissible worksheets:

$$S_{CST} = S_{step} + S_{revision} + S_{holonomy} + S_{obligation}.$$

It is not a correctness oracle. It is accepted only if invariant under admissible encoding, non-arbitrary relative to the constitutive metric and useful beyond a simpler typed graph representation.

Supplementary Note 11. Lean-facing module and theorem manifest

Local map. This Note states exactly what a Lean artifact must contain and where the trust boundary lies. It does not claim a completed kernel build in v15.

Lean is an interactive theorem prover based on dependent type theory with a small kernel that checks proof terms [23,24]. The project should use theorem parameters for external premises and avoid unregistered global axioms. Axiom dependencies must be printed for each top-level theorem; any dependency on sorryAx fails the build gate.

Proposed project structure

FmiCns/
 FSS/Basic.lean
 FSS/Semantics.lean
 FMI/Functions.lean
 FMI/SignalBoundary.lean
 Projection/Conflation.lean
 Operations/SupportSchedule.lean
 Operations/Nonreduction.lean
 Order/Composition.lean
 Order/Span.lean
 Order/Invariance.lean
 Fragmentation/Queue.lean
 Fragmentation/Regional.lean
 CNS/Lift.lean
 CNS/DynamicalBridge.lean
 Governance/Options.lean
 Propagation/Regeneration.lean
 GreatFilter/Constraints.lean
 CST/Paths.lean
 CST/Transport.lean
 CST/Closure.lean
 Main.lean

Declaration skeletons

```

structure FSS where
  Node : Type
  Edge : Type
  nodeType : Node -> TypeTag
  edgeType : Edge -> EdgeTag
  metric : Node -> Node -> Real
  incident : Edge -> Node -> Prop
  boundary : Node -> BoundaryRecord
  admissible : (Node -> Node) -> Prop
  
```

```

structure PointedFSS where
  space : FSS
  point : space.Node
  
```

```

structure Composition (X : FSS) where
  run : X.Node -> X.Node
  registered : Prop
  reusable : Prop
  spans : Prop
  
```

```

def CognitiveOrder (X : FSS) : Cardinal :=
  Cardinal.mk {c : Composition X // c.registered ∧ c.reusable ∧ c.spans}
  
```

The final implementation should quotient by functional equivalence rather than count raw implementations.

Top-level theorem signatures

```
theorem firstOrderConflation
  {W O Y : Type} (q : W -> O) (phi : W -> Y)
  {x y : W} (hq : q x = q y) (hphi : phi x != phi y) :
  Not (Exists fun decoder : O -> Y => decoder (q x) = phi x ∧ decoder (q y) = phi y) := by
  -- proof term to be supplied and kernel checked
  sorry
```

```
theorem conditionalGreatFilter
  (core : CoreCnsPremises M)
  (physical : PhysicalMapping M)
  (branch : BranchClosure M)
  (hazard : NonSummableHazard M)
  (timescale : FilterTimescale M)
  (outcome : TargetMatchingOutcome M) :
  PostIntelligenceFilter M := by
  sorry
```

The sorry markers above are documentary placeholders. A submission artifact must contain proof terms and an axiom report showing no sorryAx dependency.

Three-valued reporting outside the kernel

Lean proves propositions or fails to construct a proof. Manuscript-level reporting uses proved, disproved and undetermined. Undetermined is assigned by the workflow when neither a proposition nor its negation is certified within the declared formalization and budget; it is not a third truth value inside Lean's logic.

Supplementary Note 12. Formalization-fidelity, semantic-equivalence and mutation tests

Local map. This Note specifies the tests required before a Lean proof can be treated as a faithful formalization of the manuscript rather than a proof about a substituted object.

Formalization-fidelity audit

Independent registrars receive frozen manuscript passages and construct mappings to canonical declarations. Primary endpoints are exact recovery of semantic identity, theorem dependencies, warrant status and external-premise boundaries. Disagreement is resolved by typed counterexample rather than majority vote.

Semantic-equivalence tests

Required transformations include:

- benign node renaming;
- coordinate reparameterization preserving the metric;
- metric mutation preserving labels;
- interaction mutation preserving distances;
- warrant mutation preserving conclusions;
- participant-fitness mutation preserving aggregate mean.

Expected verdicts are invariance under the first two and semantic non-equivalence under target-relevant mutations.

Mutation catalogue

replace cognitive order with Choice depth;
 reduce Register/Recall to G/L without bridge types;
 swap support/schedule cells;
 delete a theorem premise;
 promote a conditional claim to empirical fact;
 collide aliases;
 remove a warrant edge;
 silently close an external premise;
 remove a successor obligation;
 use an intervention label as ground truth.

Every mutation must either fail type checking, invalidate a proof or produce an explicit undetermined dependency status. Diagnostic tests built from registered invariants are controls; hidden transfer is the primary evidence of propagation.

Supplementary Note 13. Executed finite computational realization

Local map. This Note preserves the previously executed finite witness with v15 warrant boundaries. It supplies constructive and conformance evidence but no external AI or civilizational measurement.

The complete state was event sourced and updated by one transition function. Each trajectory executed Register, Recall/activation, L or G with optional lift/span, certificate validation, participant Model/Evaluate/Decompose/Bridge/Governance, Store and reuse, Express and consequence realization. The final execution contained 12,800 trajectories and 102,400 events.

Deterministic witnesses

The complete model passed 23 target-pair witnesses. Each designated ablation failed or became correctly undetermined.

Record fields: ID · Contract · Target distinction · Ablation result

W01 — Contract: Register; Target distinction: external symbol becomes typed concept; Ablation result: fail
W02 — Contract: Express; Target distinction: stored verdict reconstructed externally; Ablation result: fail
W03 — Contract: Store; Target distinction: identity persists after context eviction; Ablation result: fail
W04 — Contract: Recall; Target distinction: stored bridge reactivates under later target; Ablation result: fail
W05 — Contract: L ; Target distinction: local relation classification; Ablation result: fail
W06 — Contract: G ; Target distinction: cross-region support preserves obligations; Ablation result: undetermined
W07-W09 — Contract: fitness coordinates; Target distinction: current, target and projected distinctions; Ablation result: fail
W10-W15 — Contract: internal basis; Target distinction: action-conditioned consequence and correction distinctions; Ablation result: fail / undetermined as specified
W16 — Contract: Promotion; Target distinction: promote admissible composite; Ablation result: undetermined
W17 — Contract: operand lift; Target distinction: separate lower-order-identical paired worlds; Ablation result: undetermined
W18 — Contract: certified span; Target distinction: connect lifted operand to remote target; Ablation result: fail
W19 — Contract: Certificate; Target distinction: reject decoy bridge; Ablation result: fail
W20 — Contract: Governance; Target distinction: reject mean-positive participant harm; Ablation result: fail
W21 — Contract: carrier type; Target distinction: reject signal-space regress; Ablation result: fail
W22 — Contract: target revision; Target distinction: separate target movement from progress; Ablation result: fail
W23 — Contract: one generator; Target distinction: detect split transition semantics; Ablation result: fail

Higher-order paired worlds

Record fields: Condition · Target accuracy · Certified reach · Reuse · Containment · Harm

local baseline — Target accuracy: 0.311; Certified reach: 0.000; Reuse: 0.000; Containment: 1.000; Harm: 0.000
matched compute — Target accuracy: 0.346; Certified reach: 0.000; Reuse: 0.000; Containment: 1.000; Harm: 0.000
matched memory — Target accuracy: 0.316; Certified reach: 0.000; Reuse: 0.000; Containment: 1.000; Harm: 0.000
matched population — Target accuracy: 0.333; Certified reach: 0.000; Reuse: 0.000; Containment: 1.000; Harm: 0.000

matched communication — Target accuracy: 0.400; Certified reach: 0.000; Reuse: 0.000; Containment: 1.000; Harm: 0.000
certification only — Target accuracy: 0.327; Certified reach: 0.003; Reuse: 0.000; Containment: 1.000; Harm: 0.000
lift only — Target accuracy: 0.310; Certified reach: 0.000; Reuse: 0.000; Containment: 1.000; Harm: 0.000
span only — Target accuracy: 0.317; Certified reach: 0.000; Reuse: 0.000; Containment: 1.000; Harm: 0.000
full CNS — Target accuracy: 0.585; Certified reach: 0.435; Reuse: 0.347; Containment: 0.993; Harm: 0.007
residual-open CNS — Target accuracy: 0.397; Certified reach: 0.160; Reuse: 0.130; Containment: 0.713; Harm: 0.156

Bounded registration

Record fields: Relation depth · Pipeline · Exact recovery · Relation retention · False confidence

2 — Pipeline: generic summary; Exact recovery: 0.300; Relation retention: 0.561; False confidence: 0.333
2 — Pipeline: relation extraction; Exact recovery: 0.700; Relation retention: 0.822; False confidence: 0.144
2 — Pipeline: target-aware monitor; Exact recovery: 0.867; Relation retention: 0.928; False confidence: 0.000
4 — Pipeline: target-aware monitor; Exact recovery: 0.622; Relation retention: 0.875; False confidence: 0.000
6 — Pipeline: target-aware monitor; Exact recovery: 0.233; Relation retention: 0.750; False confidence: 0.000
8 — Pipeline: target-aware monitor; Exact recovery: 0.067; Relation retention: 0.611; False confidence: 0.000
12 — Pipeline: target-aware monitor; Exact recovery: 0.000; Relation retention: 0.249; False confidence: 0.000
16 — Pipeline: target-aware monitor; Exact recovery: 0.000; Relation retention: 0.113; False confidence: 0.000

A correlation-aware retention model achieved held-out log loss 0.258 versus 1.024 for homogeneous independence. The canonical codeword reconstruction was identified at 0.940 under 1.5% independent bit flips.

v15 Gate Zero

The inherited code has not yet been audited against the expanded v15 semantic registry, order-invariance definitions or Lean manifest. No new v15 numerical claim is promoted until that audit closes.

Supplementary Note 14. Primary experiments and transfer-first propagation

Local map. This Note states the experiments that must be executed after the formal objects are frozen. Transfer and regenerative reconstruction are primary; conformance to built-in invariants is diagnostic.

EXP-GATE-ZERO-V15

Audit literal and executed definitions, label branches, surrogate metrics, actual states, registry coverage, one-generator correspondence and compatibility with the v15 semantic definitions. Failure blocks interpretation of inherited results under v15.

EXP-LEAN-FORMAL-CLOSURE

Primary outcomes: clean build, zero sorry, no unregistered axioms, complete theorem-manifest coverage, reproducible dependency report and independent build.

EXP-FORMALIZATION-FIDELITY

Independent registrars map manuscript claims to canonical definitions and theorem parameters. Primary endpoints are identity recovery, dependency recovery, warrant classification and external-premise separation.

EXP-SEMANTIC-EQUIVALENCE

Test benign and target-changing transformations from Note 12. Primary endpoints are invariance sensitivity, mutation specificity and shortest semantic witness.

EXP-LEAN-MUTATION

Run the mutation catalogue. The primary endpoint is the fraction of target-changing mutations that fail type checking or invalidate the expected top-level theorem while benign transformations preserve it.

EXP-DIRECT-RECURRENCE

Generate a first growth interval, apply a certified order lift, then continue growth under fixed order $N + 1$. The primary endpoint is a preregistered change from increasing backlog to bounded residual followed by renewed positive backlog drift.

EXP-LEAN-RECEIVER-PROPAGATION

Conditions are prose-only, registry-only, direct typed-graph formalization, CST/Lean package and corrupted controls. Receivers must recover the dependency chain, distinguish proofs from external premises, apply the formalization to hidden claim families, locate controlled mutations and produce a package from which a second receiver succeeds without author repair. Receiver and claim family are the units of inference.

EXP-CST-GEOMETRY-UTILITY

Compare direct typed graph formalization with the string/worldsheet representation. CST geometry succeeds only if it improves encoding invariance, semantic-drift detection, transfer, compression or receiver reconstruction under matched resources. Failure demotes the string-specific layer without weakening the core FSS/Lean program.

Supplementary Note 15. Registry, claim ledger and manuscript coverage

Local map. This Note specifies the source-of-truth records that prevent prose, code and formal declarations from silently diverging.

Required registries are `fss_semantics_v15`, `fmi_functions_v15`, `operation_topology_v15`, `cognitive_order_v15`, `fragmentation_dynamics_v15`, `cns_bridge_v15`, `great_filter_constraints_v15`, `cst_semantic_geometry_v15`, `lean_theorem_manifest_v15`, `claim_warrant_ledger_v15`, `experiment_registry_v15` and `version_diff_v14_to_v15`.

Every theorem record contains premises, conclusion, proof certificate or proof obligation, countermodels, dependencies, warrant, manuscript locations and Lean declaration name. Every external-premise record states its empirical interpretation, measurement protocol, status and the strongest conclusion it can promote.

Principal status ledger

Record fields: Claim · Current v15 status · Promotion requirement

Endogenous semantic identity — Current v15 status: mathematical definition and prose theorem; Promotion requirement: independent fidelity audit / executable formalization

T-CONFLATE — Current v15 status: formal proof; Promotion requirement: kernel proof term

T-ORDER-INVARIANCE — Current v15 status: prose proof under admissible-isomorphism premises; Promotion requirement: Lean quotient implementation and build

Backlog divergence — Current v15 status: formal proof; Promotion requirement: none beyond declared model

Regional fragmentation — Current v15 status: conditional theorem; Promotion requirement: explicit obligation-graph formalization

T-CNS-LIFT — Current v15 status: conditional theorem plus retained finite witness; Promotion requirement: kernel closure and direct recurrence

Dynamical bridge — Current v15 status: conditional theorem family; Promotion requirement: formal implementation of published object

Option monotonicity — Current v15 status: formal proof plus synthetic result; Promotion requirement: measured error bounds

Regenerative propagation — Current v15 status: definition; closure undetermined; Promotion requirement: independent receiver experiment

GF necessary properties — Current v15 status: decomposed conditional theorem; Promotion requirement: formal proof terms / assumption challenge

FMI-CNS structural admissibility — Current v15 status: GF-2 conditional theorem; Promotion requirement: physical and branch mapping for GF-4/5

Great Filter identification — Current v15 status: open; Promotion requirement: hazard, timescale, residue and comparative evidence

CST geometric utility — Current v15 status: open; Promotion requirement: matched comparative experiment

Lean proof closure — Current v15 status: open; Promotion requirement: pinned clean build, axiom audit, independent reproduction

Supplementary Note 16. Standalone completeness and external completion protocols

Local map. This Note is the submission gate. It states what is already contained in the package and what must still be supplied before external claims or Lean verification are promoted.

Standalone-completeness checklist

Record fields: Requirement · Disposition

Typed metric FSS and endogenous semantics — Disposition: installed
 Reflective FMI and four signal residuals — Disposition: installed
 Canonical support and schedule definitions — Disposition: installed
 Register/Recall type nonreduction — Disposition: prose proof installed
 Intrinsic order and finite certificates separated — Disposition: installed
 Representation invariance — Disposition: prose proof installed
 Backlog and regional fragmentation separated — Disposition: installed
 CNS count lemma separated from substantive lift theorem — Disposition: installed
 Bridge to published dynamical CNS — Disposition: installed conditionally
 Great Filter lemmas decomposed — Disposition: installed
 CST metric separated from optional action — Disposition: installed
 Lean module and theorem manifest — Disposition: installed as specification; build pending
 v15 Gate Zero — Disposition: pending
 direct recurrence — Disposition: pending
 independent receiver propagation — Disposition: pending
 repository and reviewer access — Disposition: pending

External completion protocols

Lean build and independent validation. Pin the toolchain, replace every documentary sorry, print axiom dependencies, reproduce in an independent environment and archive build logs.

Formalization-fidelity audit. Freeze the Article, Supplement, registry and declarations; recruit independent registrars; score identity, dependency and warrant recovery.

Receiver propagation study. Use fresh human or model receivers, hidden target families and second-receiver retransmission without author repair.

External trace study. Use version-locked systems, matched tasks, genuinely external traces and a target-aware three-valued monitor. Synthetic traces are not samples from current systems.

Historical classification. Freeze the lift rubric and anti-tuning rules before selecting candidate innovations.

Physical operation mapping. Test minimal support, factorization and commutation/precedence witnesses for the four physical assignments. Failure narrows the cross-FSS hypothesis without defeating the conceptual theorems.

Great Filter branch and hazard studies. Test branch inheritance, regenerative thresholds, hazard non-summability, correction/dispersal timescales and residue suppression.

Repository closure. Deposit code, data, registry, Lean files, environment lockfiles, checksums, source figures and manuscript source with reviewer access and a persistent DOI before publication.

The minimum submit-ready projection is the core formal article, a conformant finite witness, a clean Lean build or a clearly narrower non-Lean formal positioning, and at least one independent external anchor. The present v15 draft does not invent those missing results.

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